

DIGITAL TWIN OF AN EV POWERTRAIN

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Aim:

To develop a digital twin of an Electric Vehicle (EV) powertrain system incorporating key components such as motor, motor controller, battery, and drive cycle to simulate and monitor performance under different operational conditions.

Objective:

- Model the EV powertrain digitally using MATLAB/Simulink.
 - Simulate motor, controller, and battery across standard drive cycles.
 - Evaluate system performance (speed, torque, efficiency, SOC).
 - Develop a monitoring interface for real-time data acquisition and fault analysis.
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What We Added

In addition to the standard EV powertrain digital twin (motor, motor controller, battery, and drive cycle modeling), we advanced the system by integrating a remote controller and monitoring interface

- **Two-Way Interaction:** Simulation and hardware exchange data dynamically.
 - **Remote Command Capability:** Control acceleration, braking, or drive cycles via remote interface.
 - **Enhanced Fault Diagnosis:** Remote monitoring detects issues and enables corrective actions.
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Project Timeline

Phase 1(July - August) : Matlab model and hardware

- Vehicle dynamics and transmission model in MATLAB
- Hardware prototype (ESP32 + sensors)
- Initial closed-loop tests

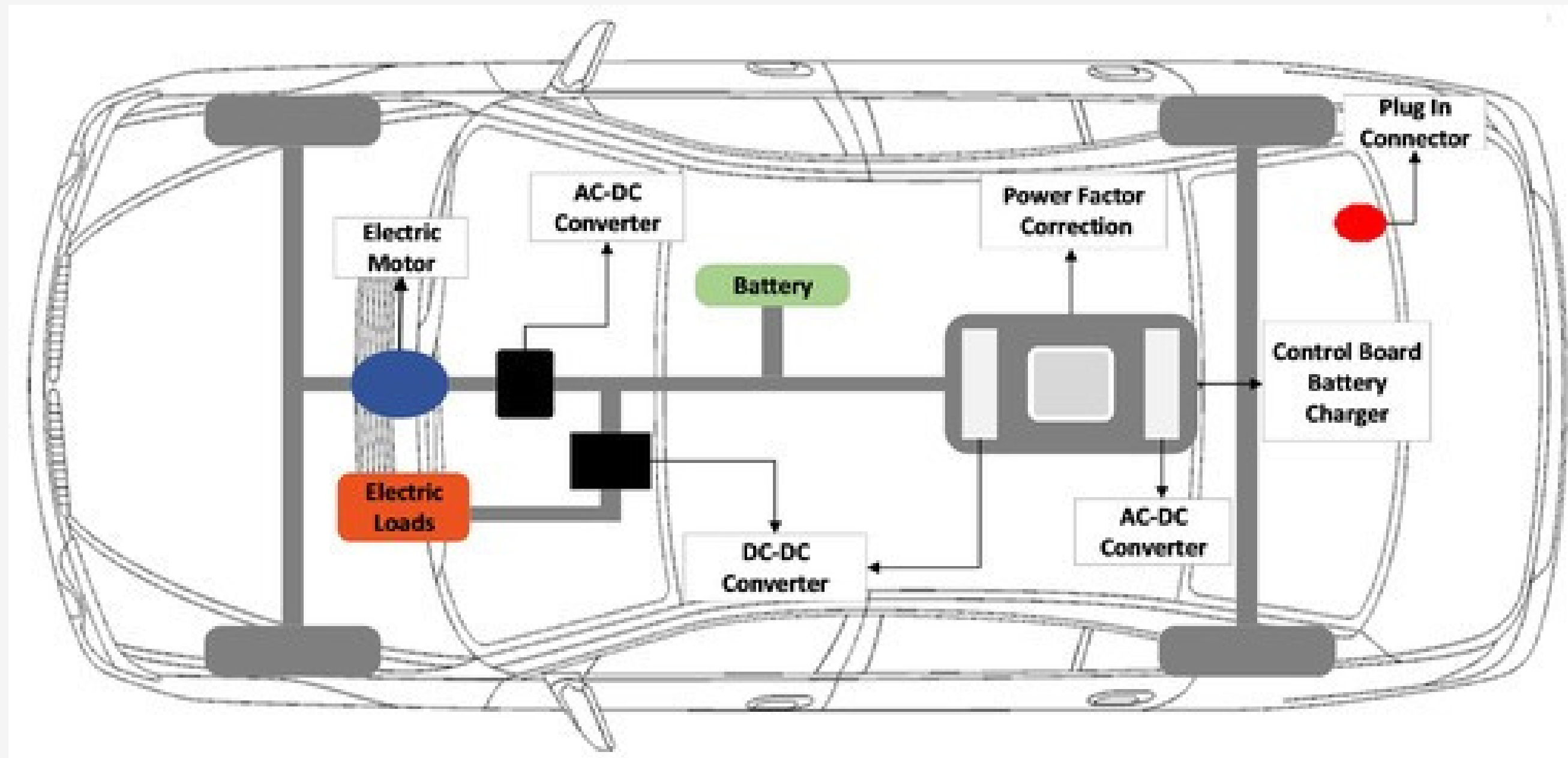
Phase 2(September-October): ADAS

- End-to-end ADAS demo
- Collision avoidance demo
- Camera + radar fusion results

Phase 3(November): autonomous

- Decision-making module
- Autonomous navigation demo
- Final integrated system validation

Main components of an EV propulsion drive system.



Components Used

- ESP32 DevKit v1 (NodeMCU-32S)
- INA219 Current/Voltage Sensor
- DS18B20 Temperature Sensor (waterproof)
- HW-477 Hall Effect Sensor (RPM measurement)
- DC Motor + Motor Driver (L298N / BTS7960)
- 12V Power Supply (adapter or Li-ion pack)
- Jumper wires + Breadboard
- USB Cable

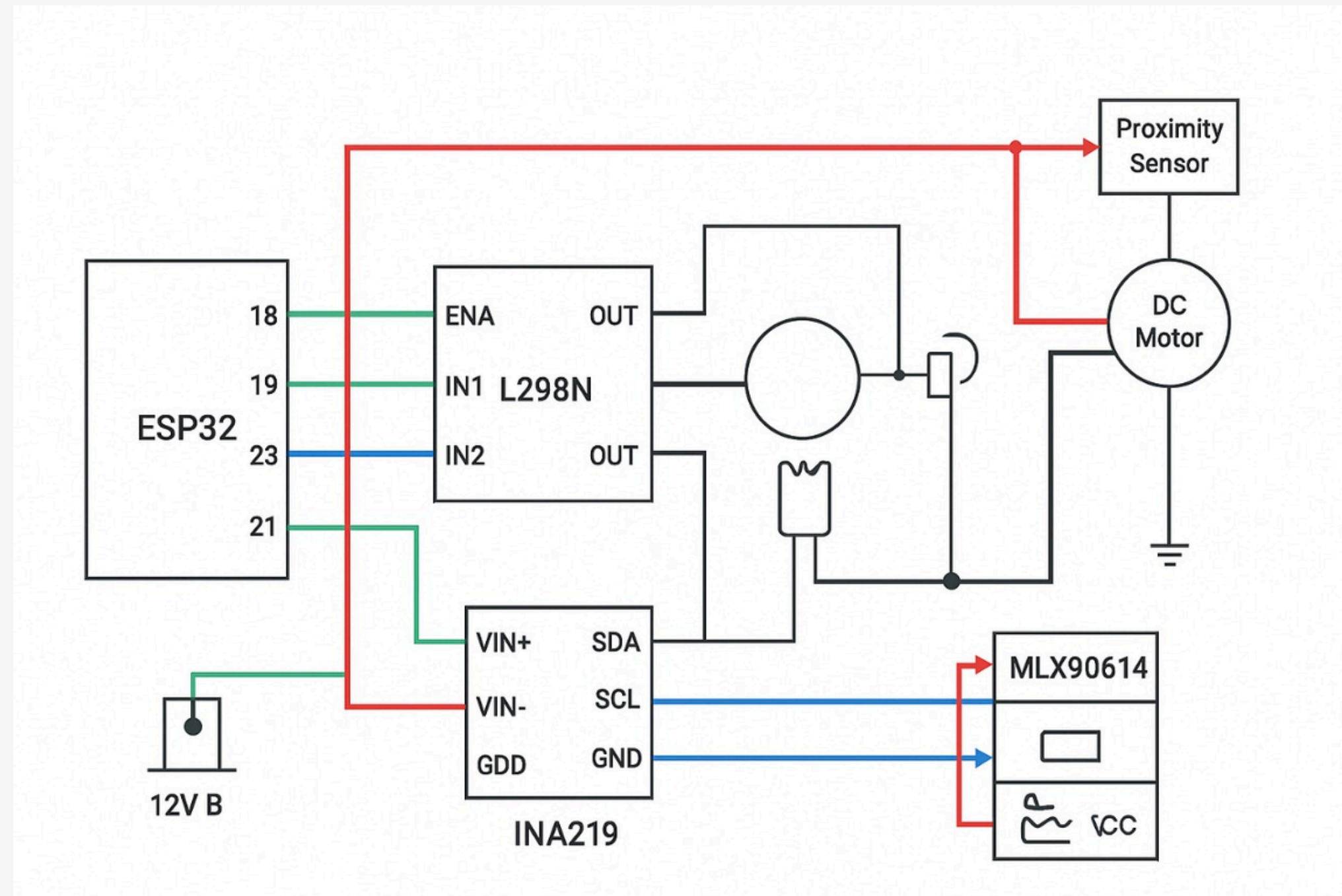


Software Tools

- **Arduino IDE:** Programming the ESP32 for sensor acquisition and Wi-Fi dashboard.
- **MATLAB:** Simulate EV drive cycles, motor dynamics, and controller behavior.
- **Simulink/Simscape:** Model electrical and mechanical subsystems (battery, motor, vehicle)
- **Relevance:** Enables two-way link between simulated twin and real-world hardware data.



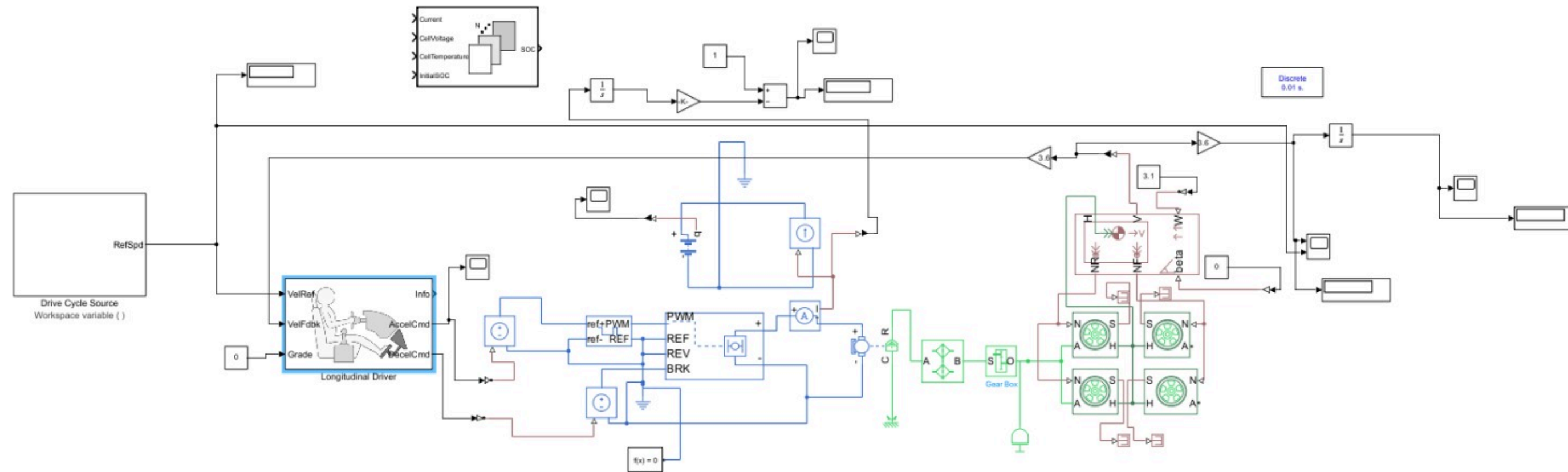
Hardware Circuit



Implementation

- **ESP32** connects to **Wi-Fi** and **streams sensor data**.
- **INA219** measures **voltage, current, power**, and **SOC%**, feeding data to the twin.
- HW-477 provides RPM pulses; DS18B20 measures motor case temperature.
- ESP32 streams JSON over serial to MATLAB for twin validation.
- **Digital twin** compares **simulated parameters** with **real-world sensor data**.

MATLAB Model



- **Input:** Standard drive cycle (velocity vs. time).
- **Processing:** Longitudinal driver block → acceleration/braking → motor controller + motor model.
- **Battery Model:** Supplies load profile under different cycles.
- **Outputs:** Motor torque, speed tracking, and SOC depletion curve.
- Sensor feedback (V, I, RPM, Temp) from ESP32 is compared against simulation results.

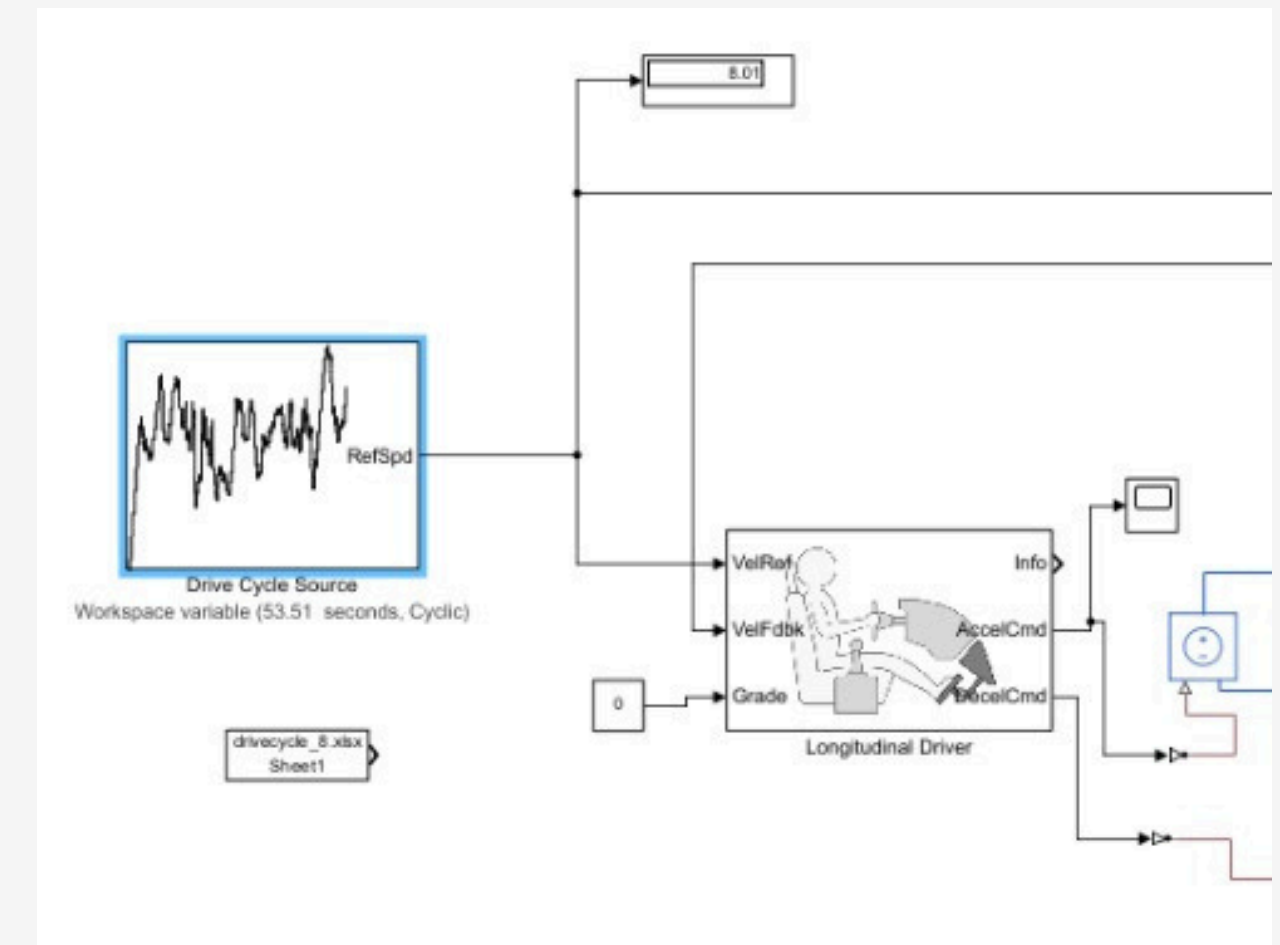
MATLAB Model

1. Drive Cycle Source (RefSpd)

- This block provides a reference vehicle speed profile (drive cycle).
- Input data comes from drivecycle_8.xlsx (Sheet1)
- Output: RefSpd – the target speed versus time.

2. Longitudinal Driver Block

- Simulates the driver's behavior (accelerator and brake control).
- Inputs:
- RefSpd → desired speed from the drive cycle.
- VelFdbk → actual vehicle speed feedback.
- Grade → road gradient (here set to 0, flat road).
- Outputs:
- AccelCmd → acceleration (throttle) command.
- DecelCmd → braking command.



MATLAB Model

1. Battery (Top-Left Block):

- Inputs: current, cell voltage, cell temperature, initial SOC.
- Output: SOC (State of Charge) of the battery.
- Tracks battery status during the drive cycle.

2. Motor Controller (Middle Section):

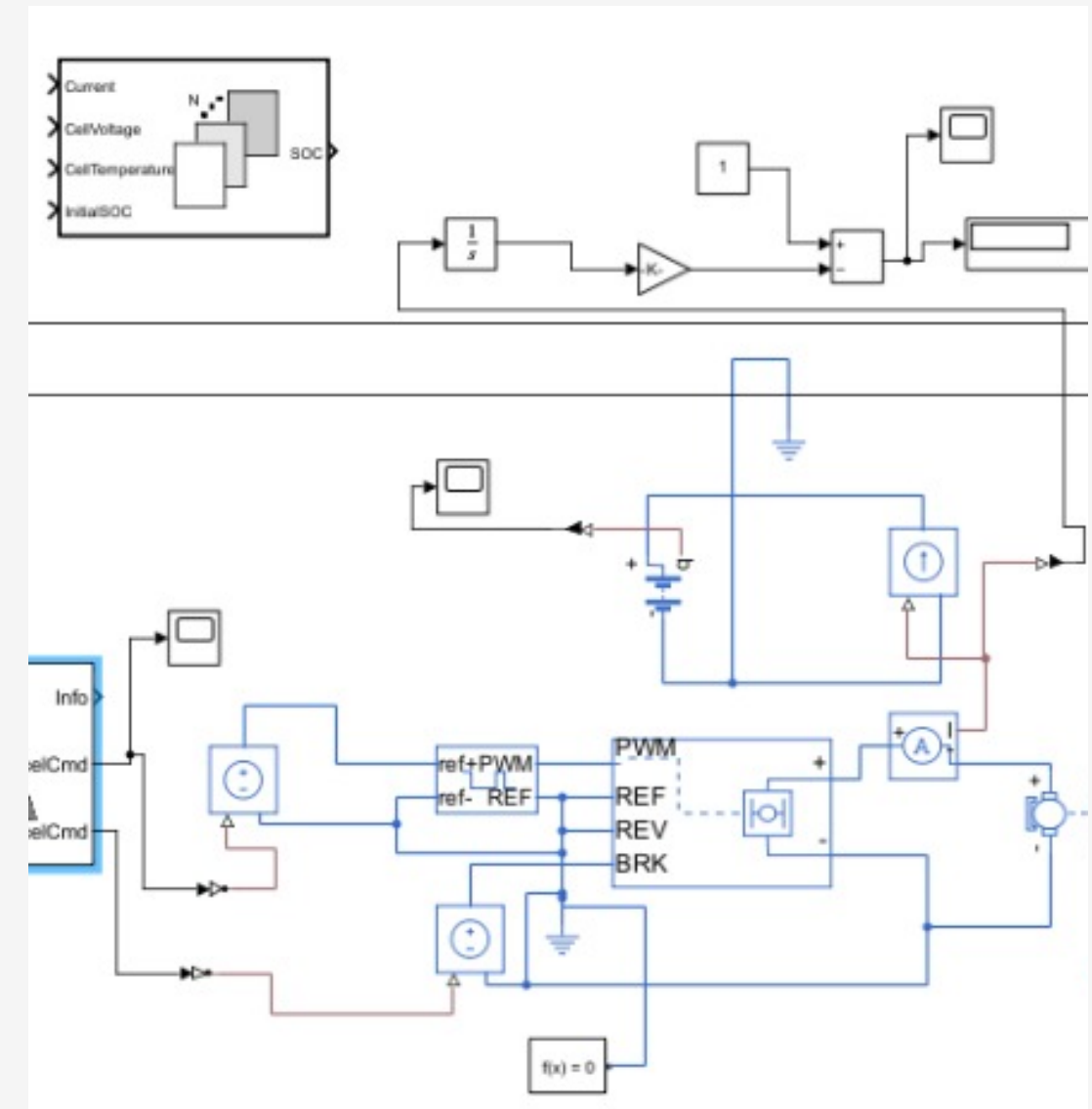
- Inputs: AccelCmd and DecelCmd from the driver.
- Generates PWM, REF, REV, BRK signals → used to control the motor.
- Acts as an inverter/controller to regulate motor torque and direction.

3. Electric Motor (Right Side, Circle with Shaft):

- Converts electrical input (PWM from controller) into mechanical rotation.
- Provides torque and speed to the drivetrain.

4. Feedback Loops (Blue Lines):

- Current and voltage feedback to controller → ensures correct motor operation.
- SOC feedback → ensures energy management is consistent.



MATLAB Model

1. Inverter (Bottom-Left, Green Section):

- Converts DC from the battery into AC using power electronics.
- Uses a switching bridge (with IGBTs/MOSFETs) to generate 3-phase AC.

2. 3-Phase Motor (Center, Four Coils in Square):

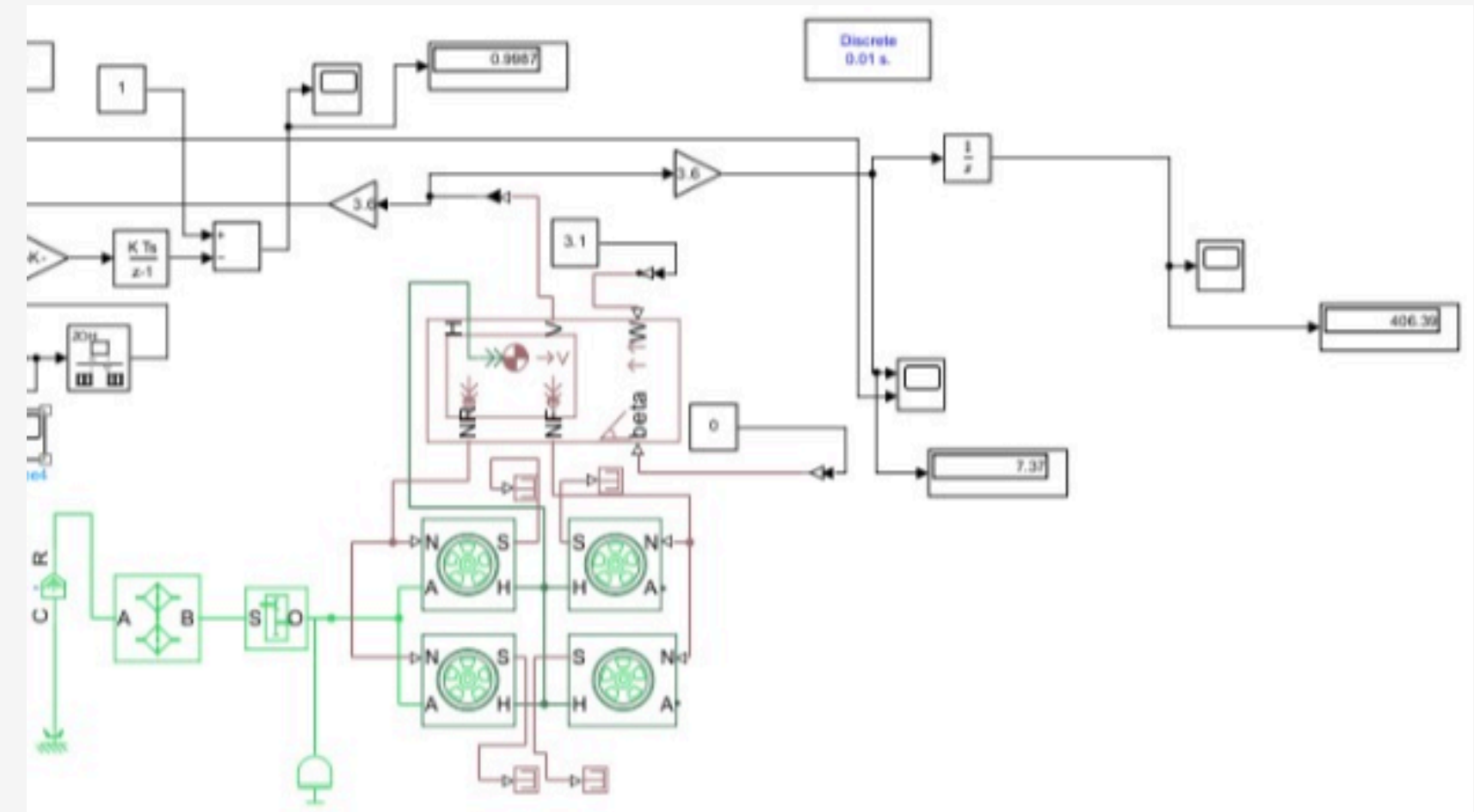
- This is the three-phase induction/PMSM motor model.
- Inputs: 3-phase AC from inverter.
- Output: Mechanical torque and rotor speed.

3. Control Signals (Top Section):

- PWM and reference signals regulate inverter switching.
- Feedback loops (speed, torque, and current) ensure motor follows the driver's commands.

4. Measurement Blocks (Right Side):

- Output displays torque (~406 Nm) and speed (~7.35 rad/s).
- Used to validate motor performance under the given drive cycle.



Mathematical Modelling

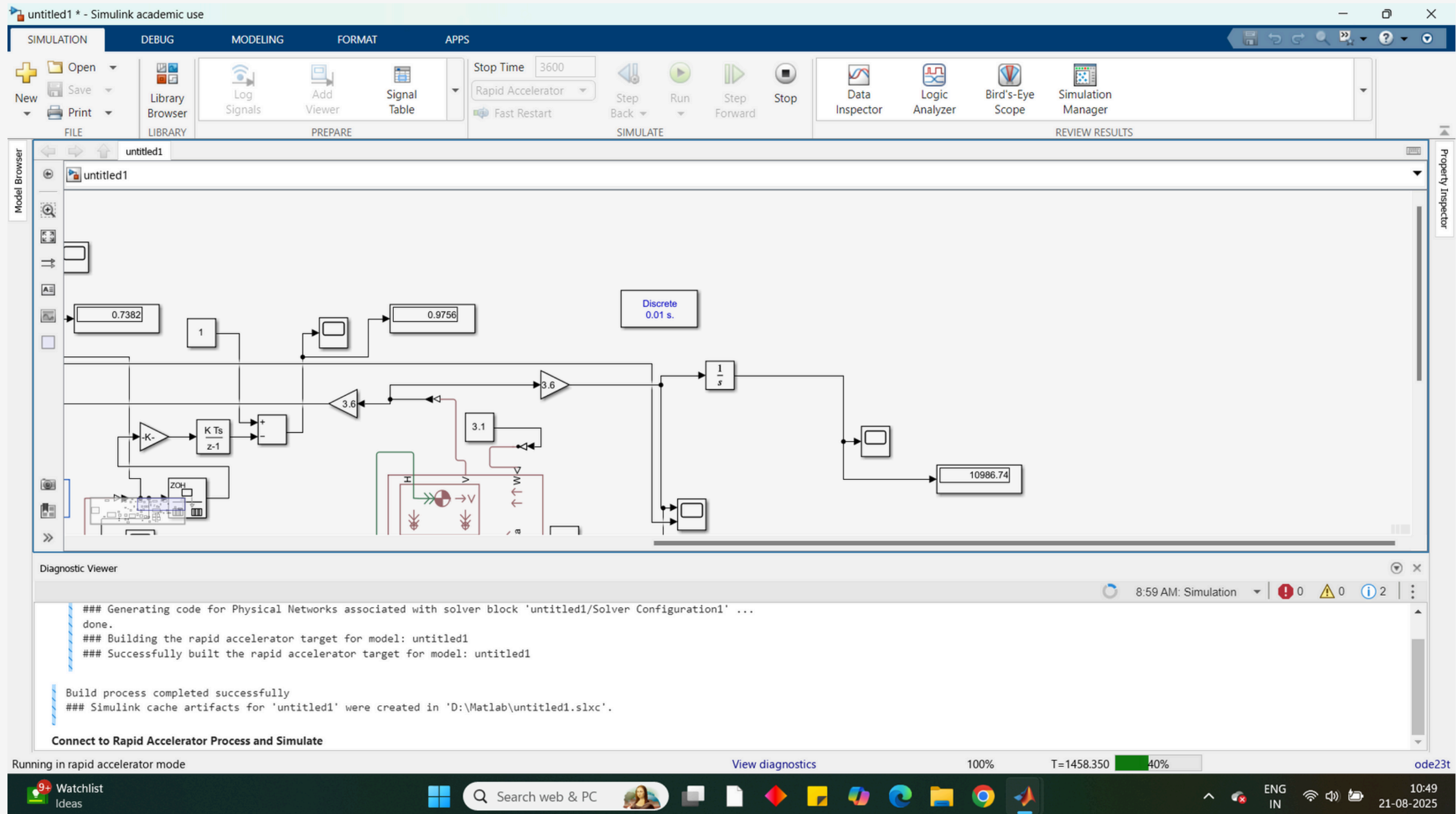
- Battery SOC estimation:

$$SoC(t) = SoC(t - 1) + \frac{I(t)}{Q_n} \Delta t$$

- Drive cycle modeling:
 1. Velocity vs. Time curve.
 2. Torque demand
 3. Motor model



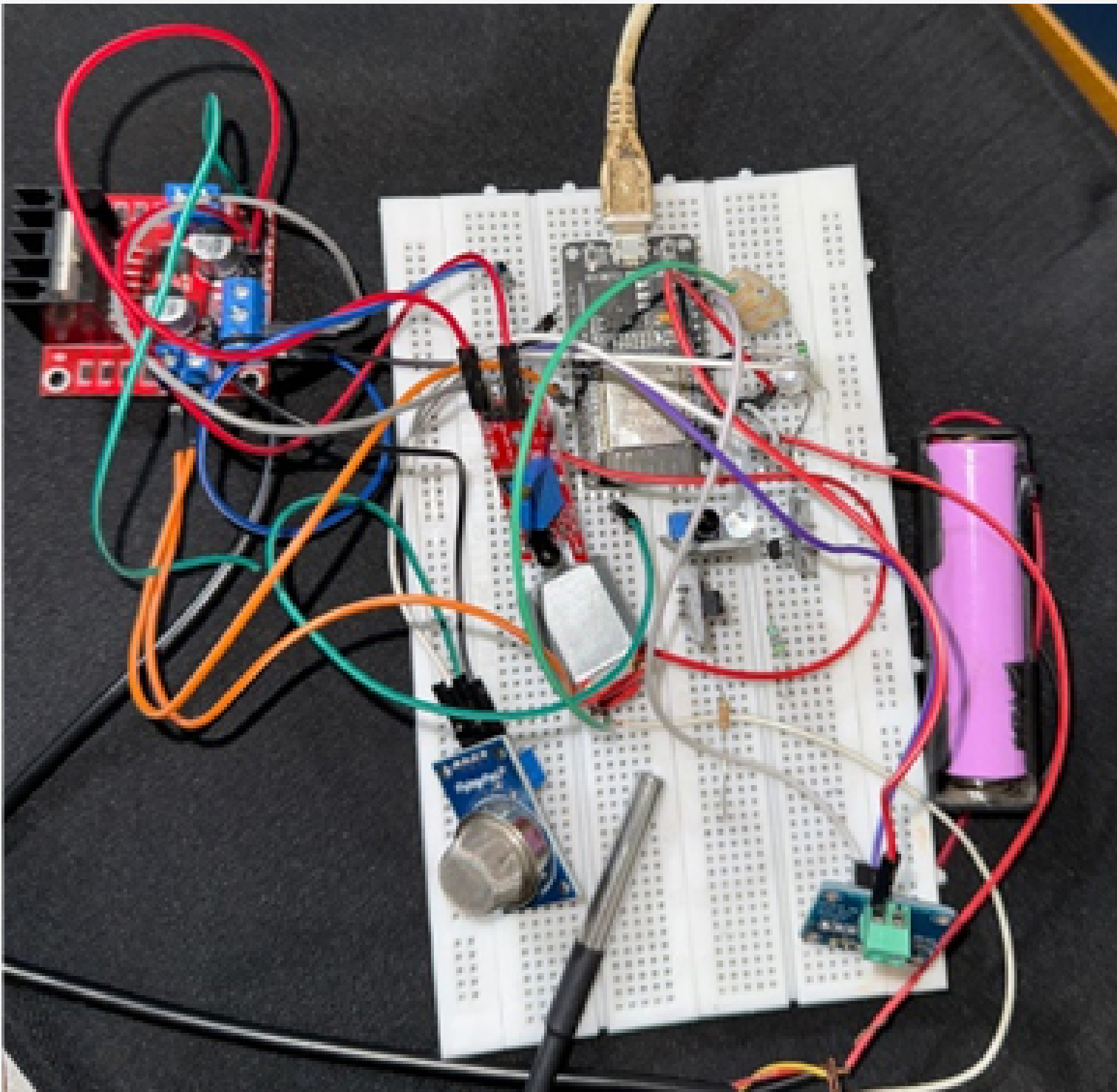
Simulation Results



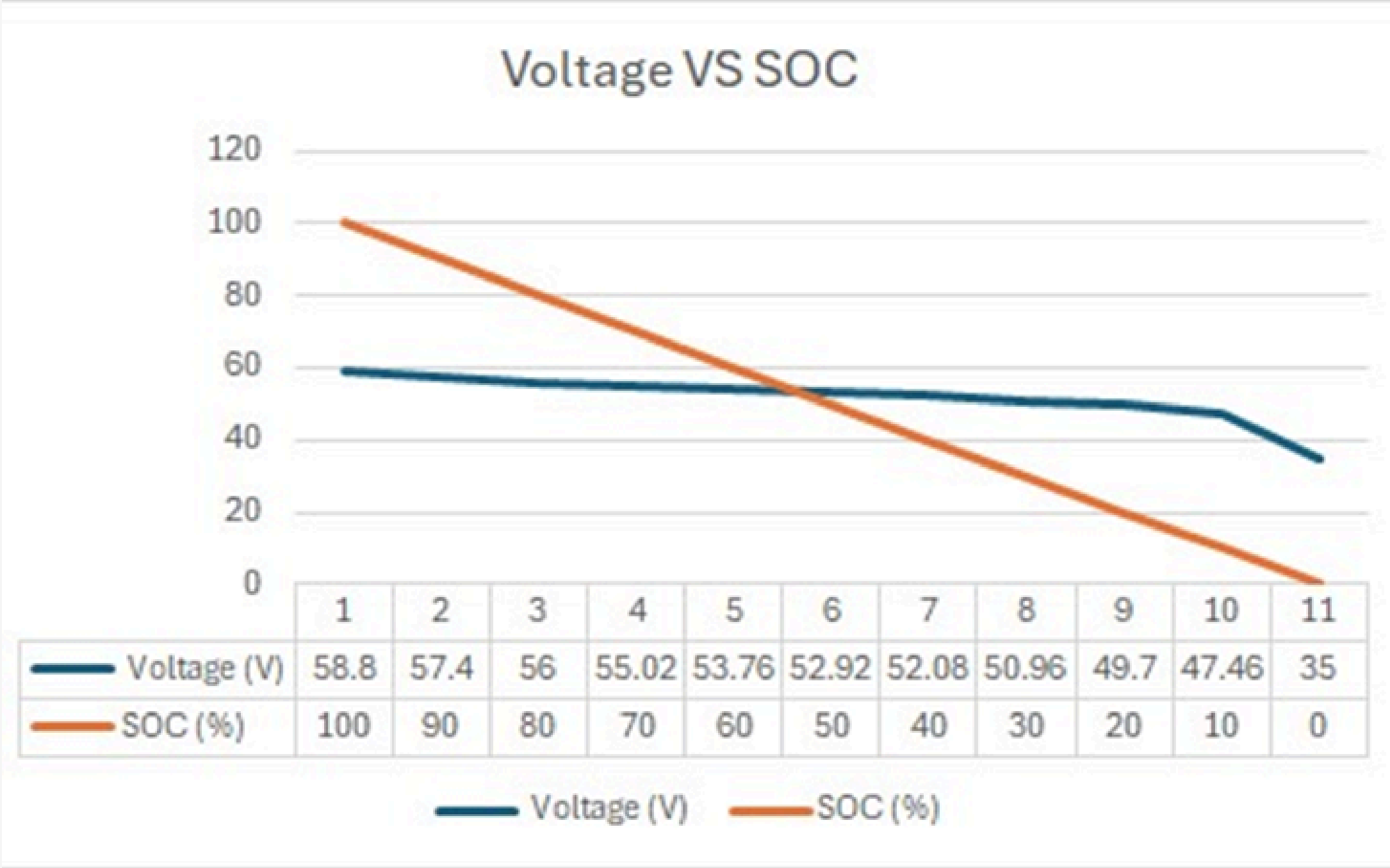
Sensor Output

COM5

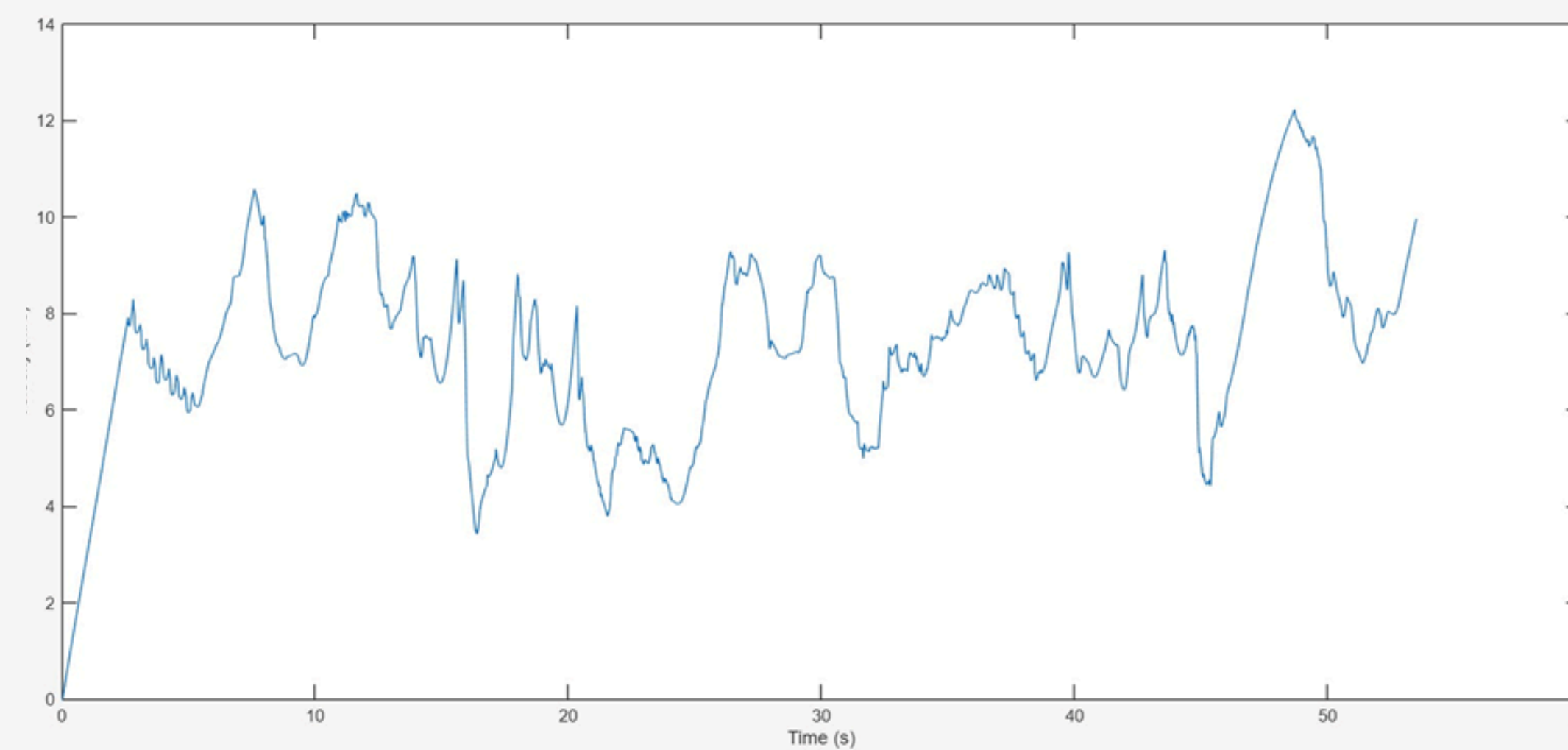
```
23:49:25.645 -> Shunt Voltage:      0.18 mV
23:49:25.645 -> LKD Current:      1.90 mA
23:49:25.645 -> Power Consumption: 6.00 mW
23:49:25.645 -> Battery Level:      73.35 %
23:49:25.645 -> Battery Status:    Good
23:49:25.645 -> LKD Status:      ON - Low Current
23:49:25.645 -> Approx. Resistance: 979.04 Ω
23:49:25.645 -> Estimated Runtime: 40.2 days
23:49:25.645 ->
23:49:25.645 -> -----
23:49:25.645 ->
23:49:27.650 -> 10650 Battery & LKD Load Readings:
23:49:27.650 -> -----
23:49:27.650 -> Battery Voltage (load): 3.06 V
23:49:27.650 -> Bus Voltage:      3.06 V
23:49:27.650 -> Shunt Voltage:      0.16 mV
23:49:27.650 -> LKD Current:      1.80 mA
23:49:27.697 -> Power Consumption: 6.00 mW
23:49:27.697 -> Battery Level:      72.91 %
23:49:27.697 -> Battery Status:    Good
23:49:27.697 -> LKD Status:      ON - Low Current
23:49:27.697 -> Approx. Resistance: 1031.20 Ω
23:49:27.697 -> Estimated Runtime: 42.2 days
23:49:27.697 ->
23:49:27.697 -> -----
23:49:27.697 ->
```



Data Flow from Physical Vehicle to Digital Twin

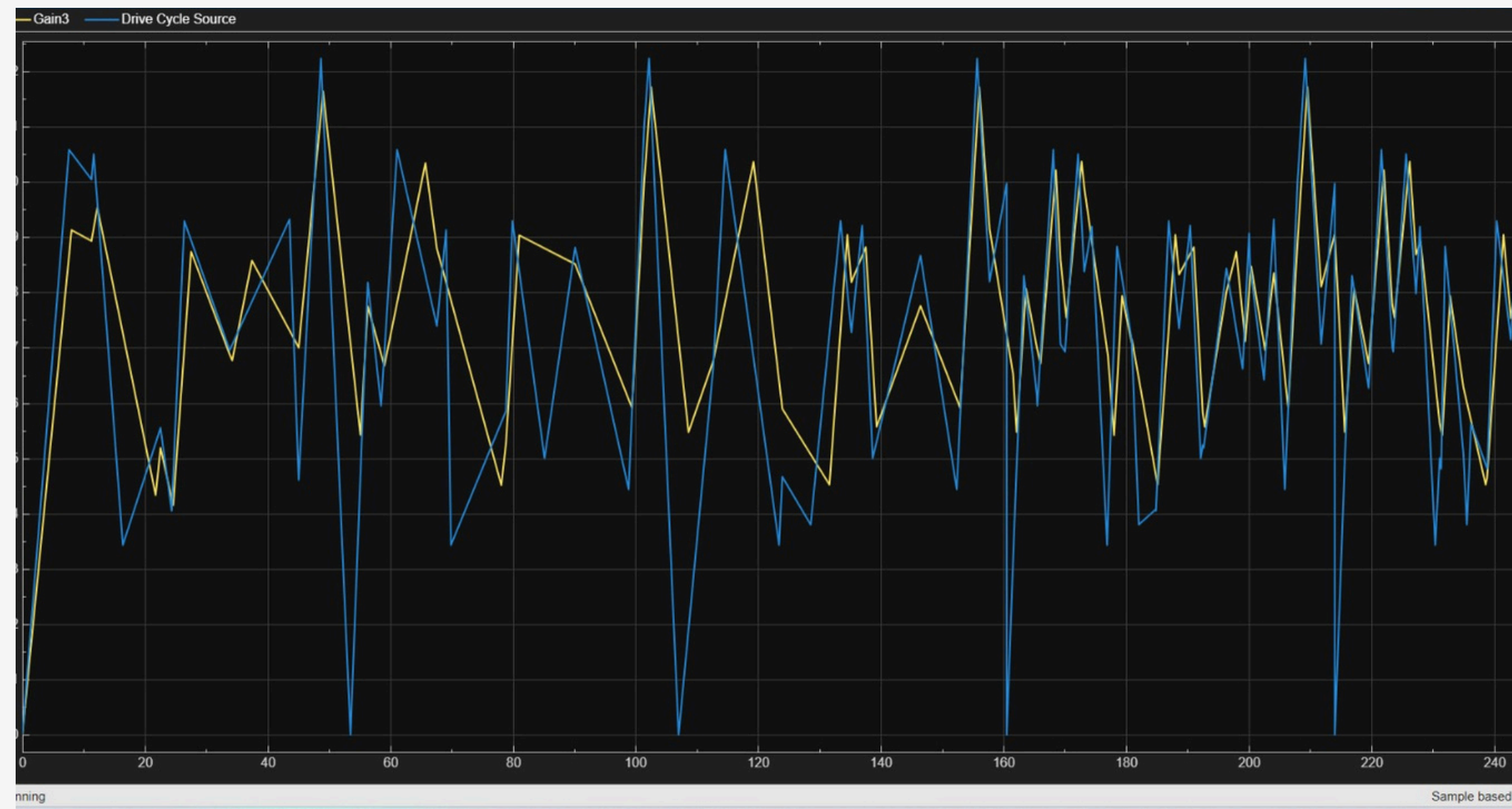


Voltage vs SOC



Velocity vs Time

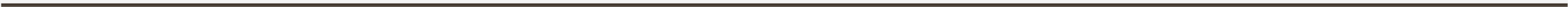
**Real Speed vs
Simulated Speed**



References

1. Grieves, M. (2014). Digital Twin: Manufacturing Excellence through Virtual Factory Replication.
2. MATLAB Documentation – Electric Vehicle Modeling with Simscape Electrical, MathWorks, 2024.
3. Li, Y. et al. (2020). Digital Twin-Driven Smart Manufacturing. Robotics and Computer-Integrated Manufacturing, 61, 101837.
4. IEC 61850-90-3: Utilization of IEC 61850 for Electric Vehicle Charging Systems.
5. Singh, P. et al. (2022). Development of Real-Time Electric Vehicle Simulation using MATLAB/Simulink. International Journal of Vehicle Engineering, Vol. 14, No. 3.





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THANK YOU

